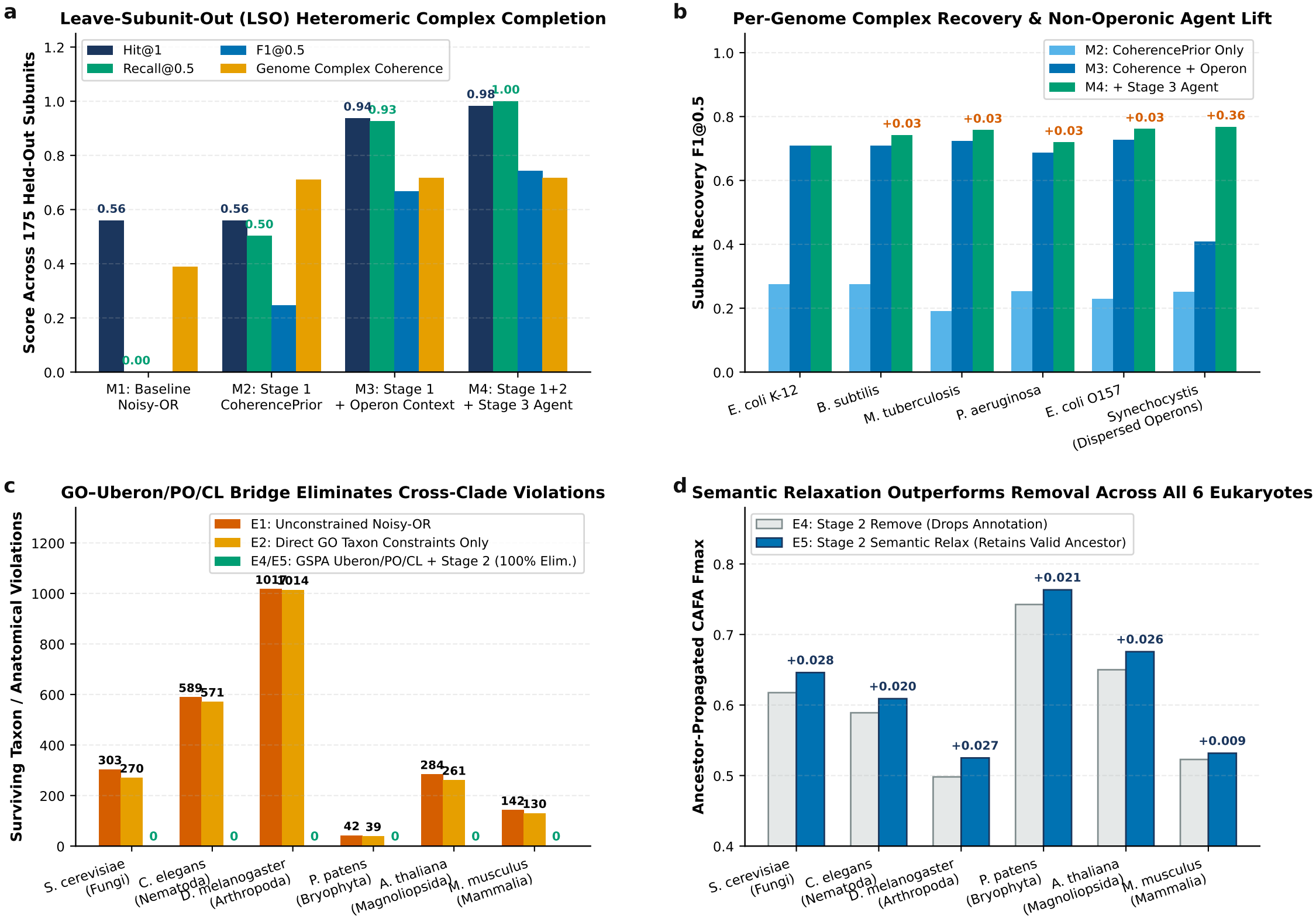
a
Leave-Subunit-Out (LSO) Heteromeric Complex Completion
Score Across 175 Held-Out Subunits
Hit@1
F1@0.5
Recall@0.5
Genome Complex Coherence
0.56
0.00
0.56
0.50
0.94
0.93
0.98
1.00
M1: Baseline Noisy-OR
M2: Stage 1 CoherencePrior
M3: Stage 1 + Operon Context
M4: Stage 1+2 + Stage 3 Agent
b
Per-Genome Complex Recovery & Non-Operonic Agent Lift
Subunit Recovery F1@0.5
M2: CoherencePrior Only
M3: Coherence + Operon
M4: + Stage 3 Agent
+0.03
+0.03
+0.03
+0.03
+0.36
E. coli K-12
B. subtilis
M. tuberculosis
P. aeruginosa
E. coli O157
Synechocystis (Dispersed Operons)
c
GO-Uberon/PO/CL Bridge Eliminates Cross-Clade Violations
Surviving Taxon / Anatomical Violations
E1: Unconstrained Noisy-OR
E2: Direct GO Taxon Constraints Only
E4/E5: GSPA Uberon/PO/CL + Stage 2 (100% Elim.)
303
270
0
589
571
0
1011
1014
0
42
39
0
284
261
0
142
130
0
S. cerevisiae (Fungi)
C. elegans (Nematoda)
D. melanogaster (Arthropoda)
P. patens (Bryophyta)
A. thaliana (Magnoliopsida)
M. musculus (Mammalia)
d
Semantic Relaxation Outperforms Removal Across All 6 Eukaryotes
Ancestor-Propagated CAFA Fmax
E4: Stage 2 Remove (Drops Annotation)
E5: Stage 2 Semantic Relax (Retains Valid Ancestor)
+0.028
+0.020
+0.027
+0.021
+0.026
+0.009
S. cerevisiae (Fungi)
C. elegans (Nematoda)
D. melanogaster (Arthropoda)
P. patens (Bryophyta)
A. thaliana (Magnoliopsida)
M. musculus (Mammalia)